

MINDTREK

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Sovereign *by habit*

20 years of self-hosting from source
on European servers

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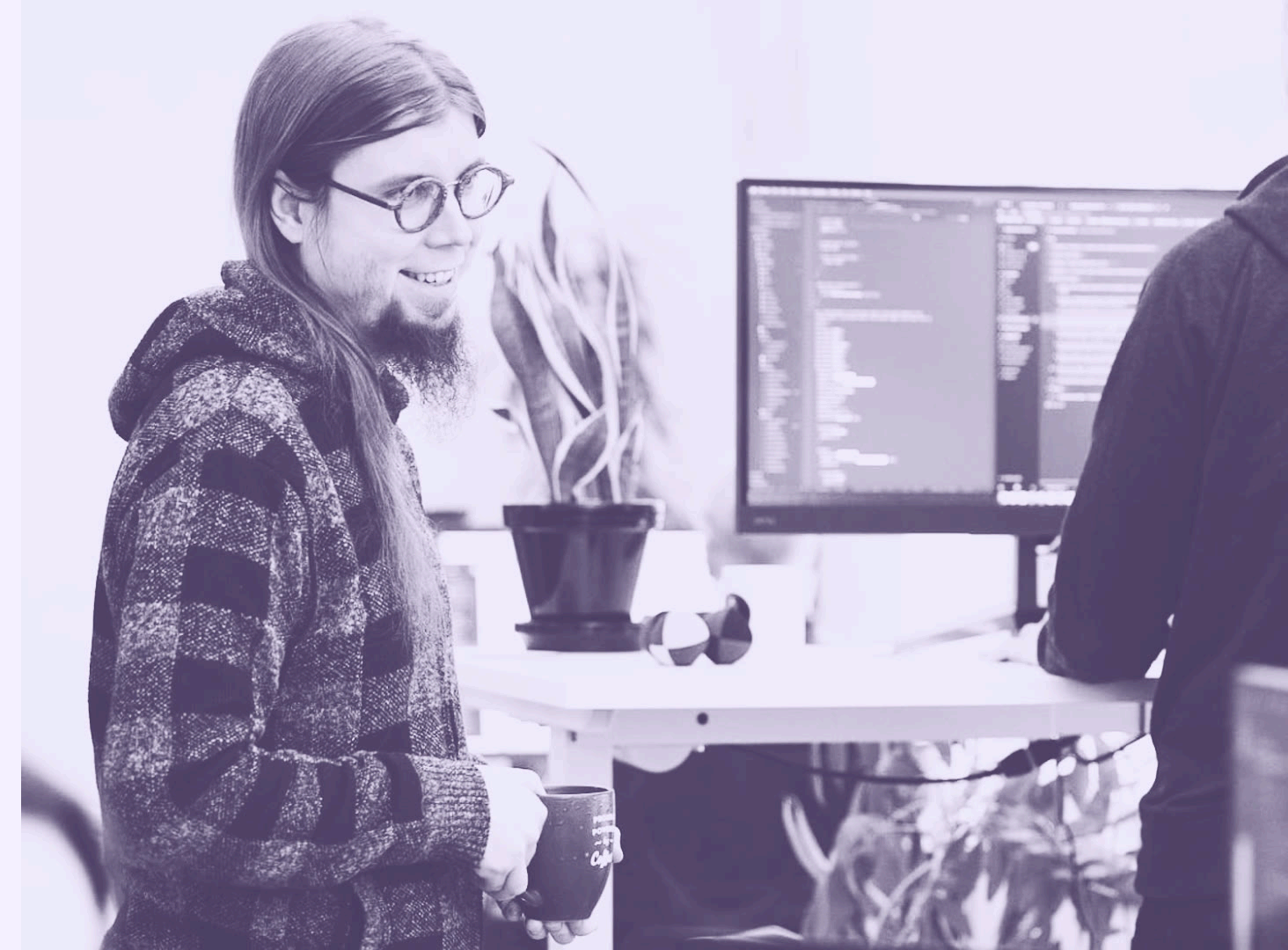


About *Rolle*

Founder and CTO of Digitoimisto Dude Oy. Code soon 30 years, WordPress since 2005, online since 1999. Accessibility, open source, Linux servers, and building my own tools. Free time: running, bilingual family, knows sign language.

Dude & *WordPress*

Contributing to WordPress and open source since we founded Dude in 2013: 71 open source repositories on GitHub, including air-light, our starter theme with over 1100 stars.



Sovereignty as a strategy, *or as a habit*

Europe's digital sovereignty has become a strategy topic,
something organisations plan their way towards.
This talk is about getting there by habit.

Layers we rarely *look inside*

Containers

An image bundles the app with an operating system someone else chose.

Orchestration

A scheduler decides where and when the containers run.

Managed platforms

No shell and no filesystem. You deploy, the platform runs it.

Serverless

Functions run on demand in another company's runtime.

Where the habit *comes from*

1996

KDE at 8 years old, on the family's Red Hat based Linux.

School

An IRC shell account on TNNet.

1998

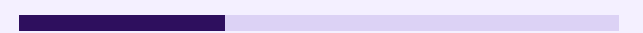
Mandrake Linux, dual-booted on my sister's PC.

Later

A headless server in the wardrobe, used over SSH.

Dude 2013-2015: *from partners to our own server*

- 2013: client sites on hosting partners' servers.
- Jails, cPanel and control panels stood between us and the server.
- 2015: our first own VPS, to run the latest nginx.
- Apache out, nginx in, for good.



Choosing providers, *and leaving them*

- 2015 DigitalOcean: the first VPS
- 2016 OVH in France: 2 web, 2 database and 1 file server
- 2016 The outage: every client site hangs
- 2016 Every site moved to Multim in Pori in one night
- 2016-2026 A datacentre in a former army rock cave, on wind power
- 2026 Moving everything to one Finnish provider



The fleet *today*

About 30 servers

Ubuntu 24.04 on almost all of them.

One stack per server

nginx, PHP-FPM, MariaDB, Valkey and fail2ban on the same machine.

15 people

A web agency with no separate operations team.

Our own services

Analytics, docs, CRM and a Mastodon instance with 1000 users.

From source *and from upstream*

Built from source

nginx modules for Brotli, cache purging and GeoIP, against the exact nginx that runs.

Distribution packages

Everything else comes from Ubuntu.

Upstream repositories

nginx from nginx.org, PHP from the Ondřej Surý PPA, MariaDB from MariaDB.

Configured by hand

Every configuration file is ours, with our own defaults.

When a layer *breaks*

Read the running build

```
nginx -V 2>&1 | grep configure  
apt-get source nginx
```

Build the module against it

```
cd nginx-*/  
./configure <same arguments> \  
    --add-dynamic-module=../ngx_cache_purge  
make modules
```

Ubuntu patched nginx for a security fix, and our modules built from plain nginx source started crashing. Building them against Ubuntu's own source, with the same configure arguments, fixed it.

Automation, *and what stays manual*

Scripts

Bash install and maintenance scripts since 2025, now moving to Ansible.

Monitoring

Logs, errors and suspicious requests are watched and reported to Slack.

Every hour

Database and file backups to a storage box, with daily and weekly snapshots.

By hand

Monthly patching follows a written runbook and takes about 3 hours.

Where it *costs more*

- Monthly patching takes hours, and only two people can do it.
- Servers set up at different times drift apart.
- Fixes are made by hand first and written into scripts afterwards.
- Knowledge stays with a few people unless it is written down.

Where it *pays*

- Every layer can be read, so nothing is a black box.
- A crash can be traced from the log to the source.
- In 2016 we moved every site to another provider in one night.
- The 2026 move to one provider is planned in waves over three months.



Where it **still** *depends on others*

DNS and CDN

Cloudflare, for DNS, caching and the Mastodon media storage.

Code and tools

GitHub for code, Better Stack for uptime, Let's Encrypt for certificates.

Email

Mailgun sends it, and delivery depends on the large mail providers.

AI

Log analysis uses OpenAI models, and I use Claude a lot.

Knowing every layer *is practical sovereignty*

Host one thing yourself, end to end,
and keep it running for a year.

Kiitos. *Questions?*

Rolle Laukkarinen

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